Cryptography from Sampling Oracles and LOCC Primitives in the Bounded CRQS Model

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Ryo Nishimaki, Nick Spooner, Takashi Yamakawa

EPFL NTT Cornell University

[BFM88; MNY24; Qian24]

CRS

The common reference string model

[BFM88; MNY24; Qian24]

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public algorithm: g

[BFM88; MNY24; Qian24]

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CRS $\leftarrow g(k)$

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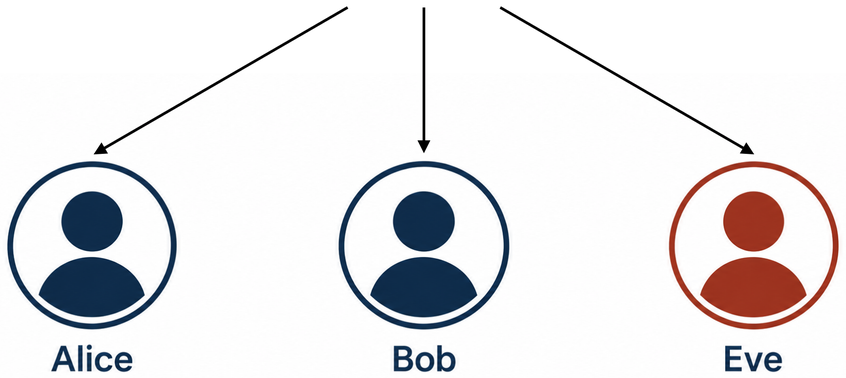
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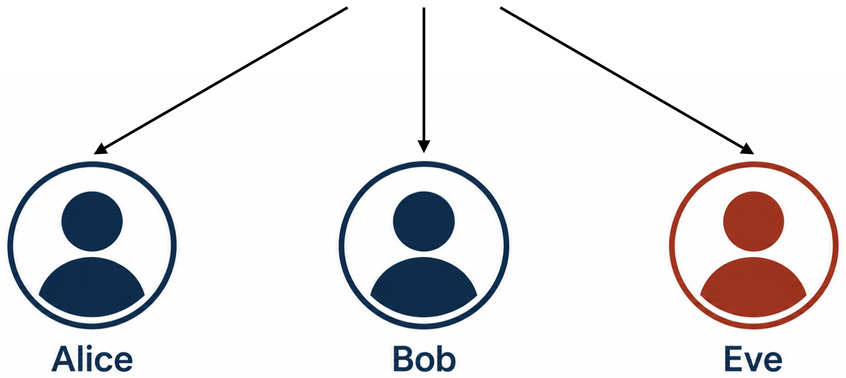
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Give constructions impossible in plain model:

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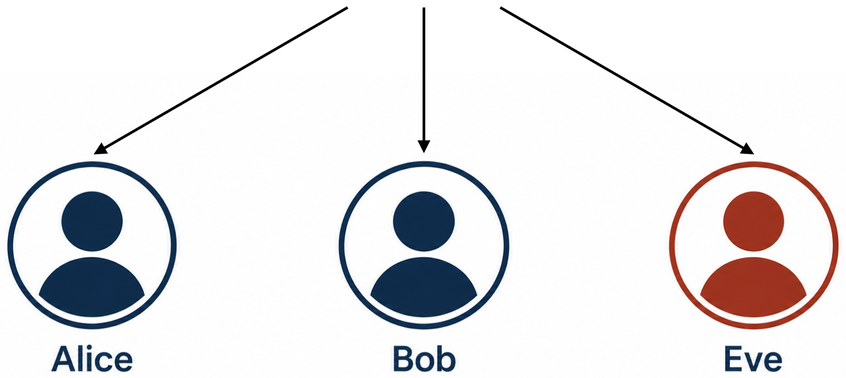
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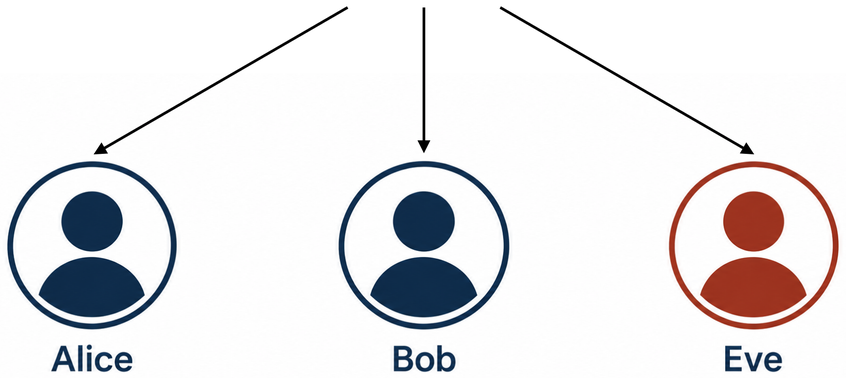
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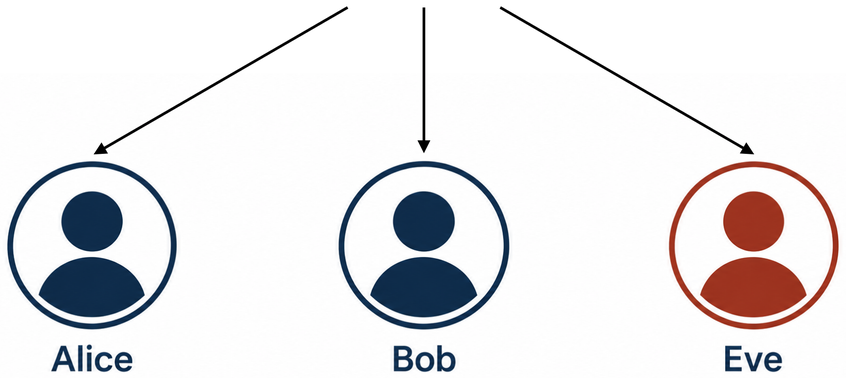
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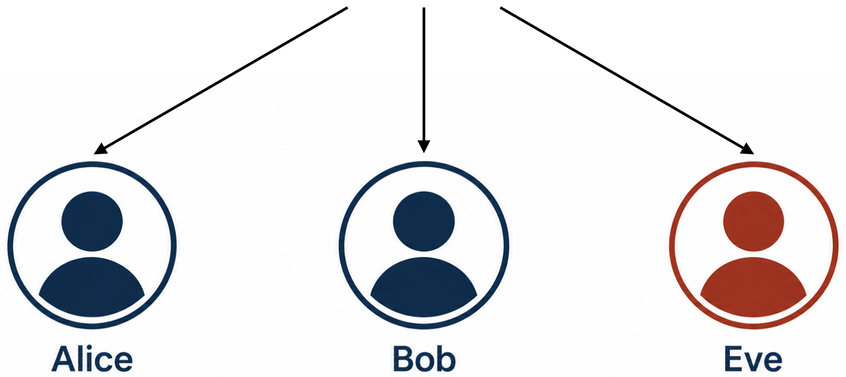
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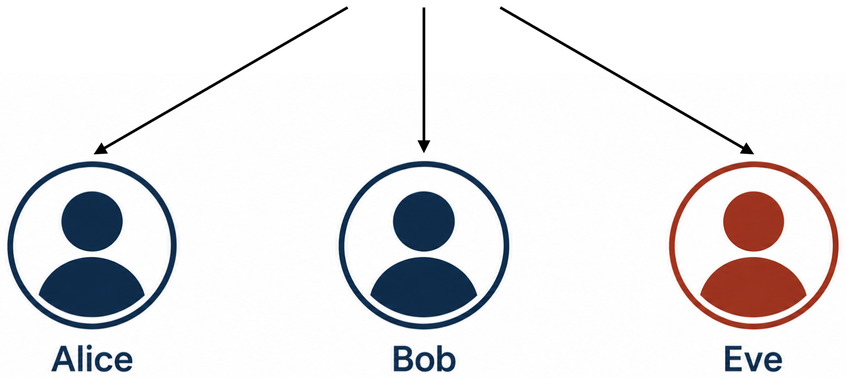
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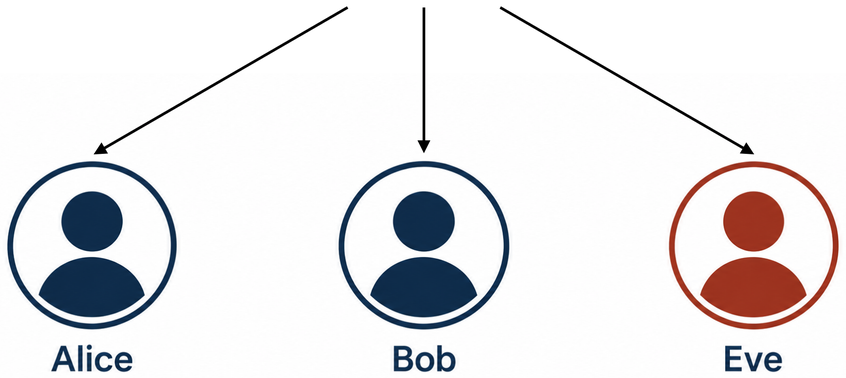
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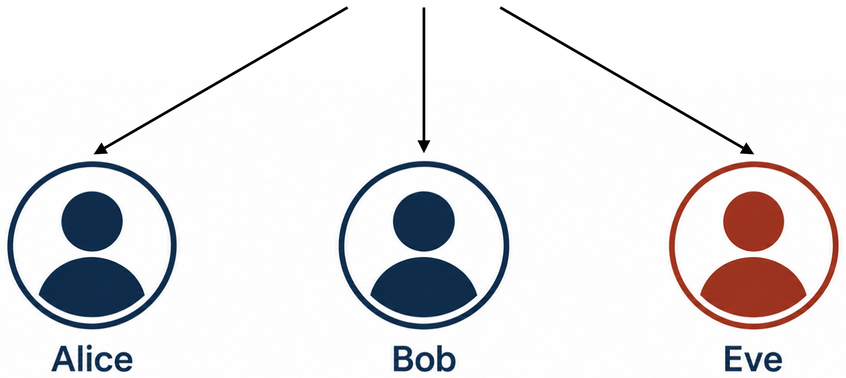
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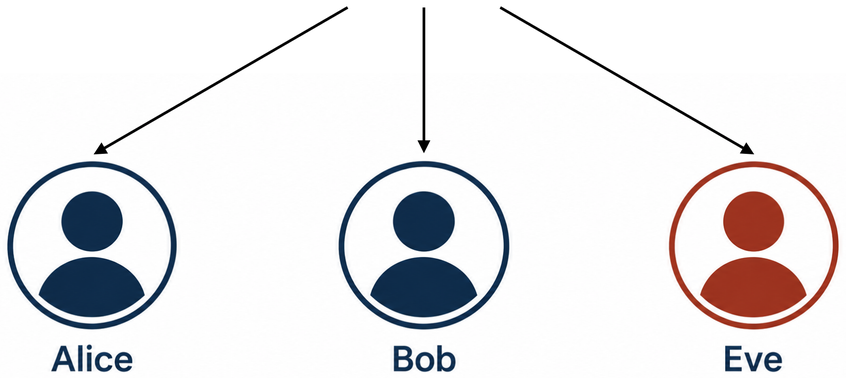
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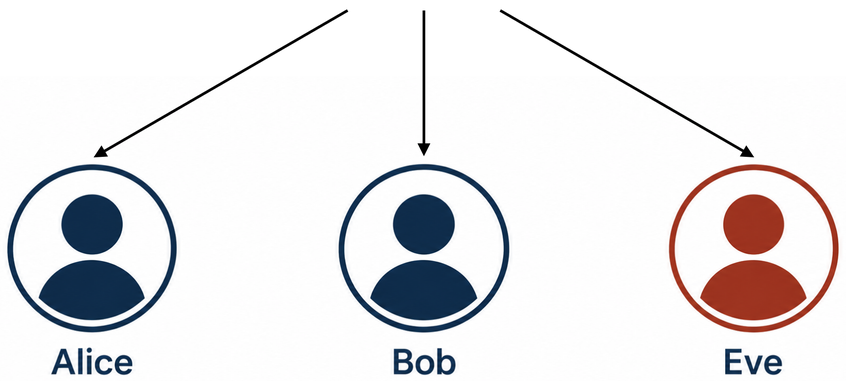
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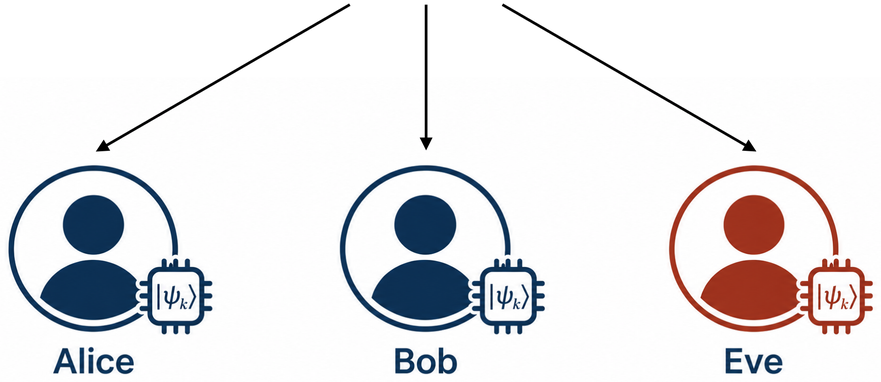
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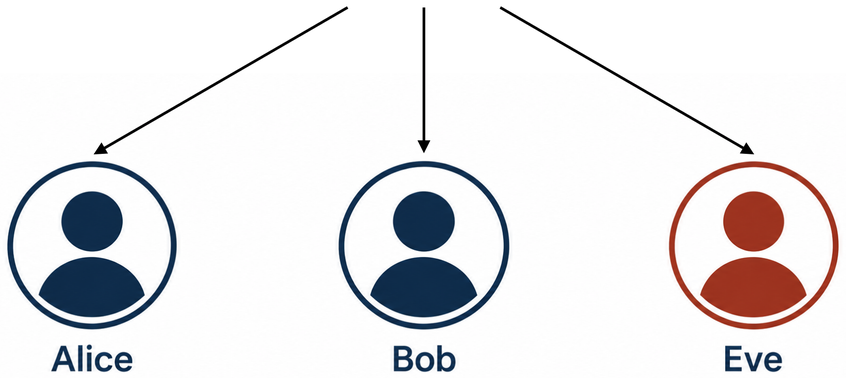
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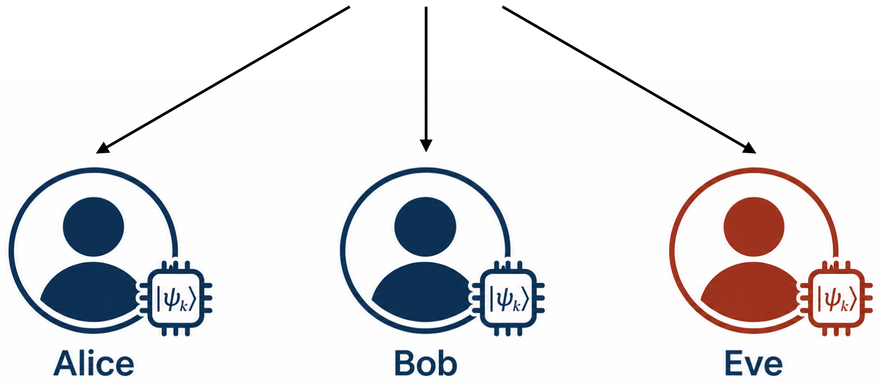
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More constructions:

Statistically-secure (hiding,binding) quantum commitments in bounded setting

[BFM88; MNY24; Qian24]

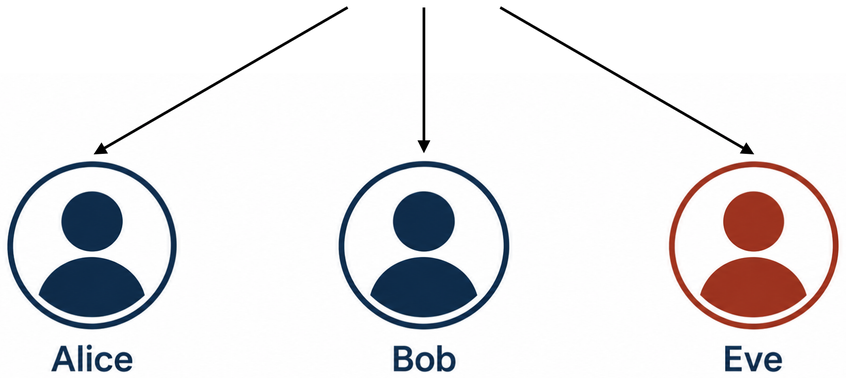
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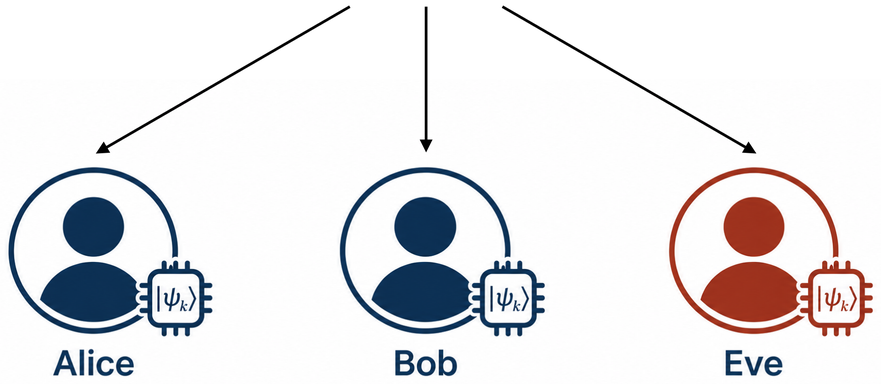
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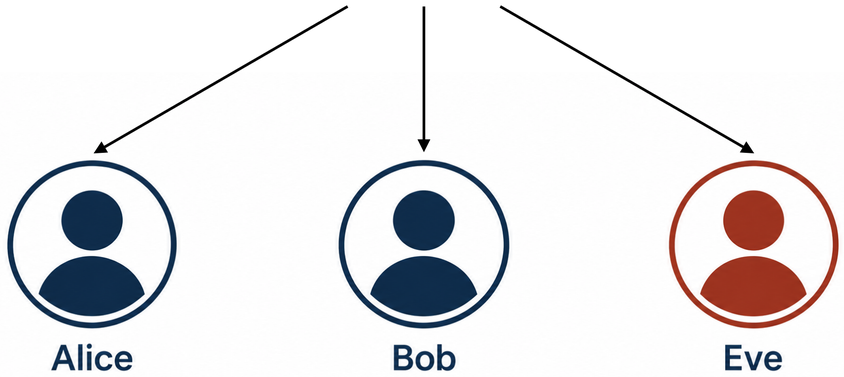
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Pure quantum state

Bounded - adversary receives *a priori* t copies
Unbounded - adversary receives any polynomial number of copies.

More constructions:

Statistically-secure (hiding,binding) quantum commitments in **bounded** setting

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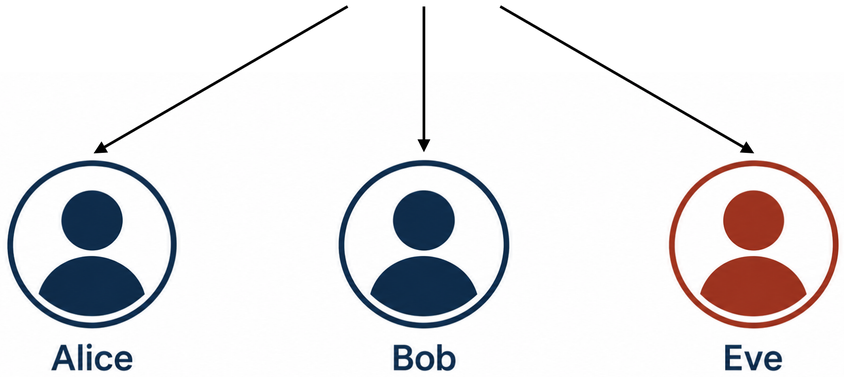
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public quantum algorithm: Q

$k \leftarrow \{0,1\}^k$

$|\psi_k\rangle \leftarrow Q(k)$ — Pure quantum state

Our focus

Bounded - adversary receives *a priori* t copies

Unbounded - adversary receives any polynomial number of copies.

More constructions:

Statistically-secure (hiding,binding) quantum commitments in **bounded** setting

Impossible in CRS

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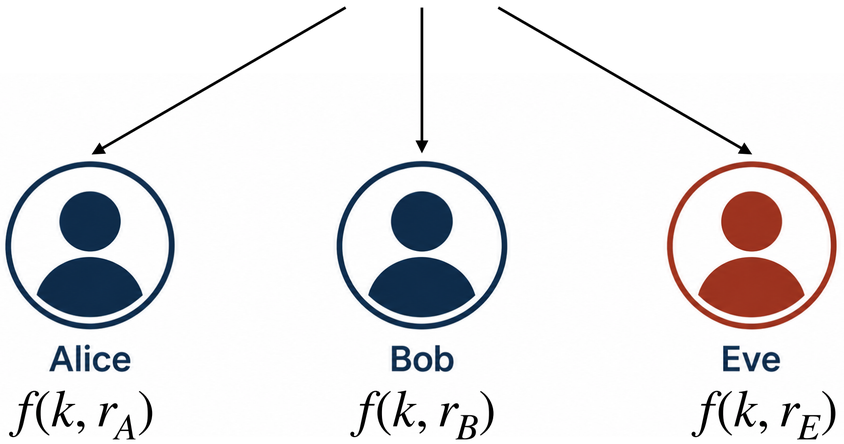
The common reference distribution model

Compared with CRS

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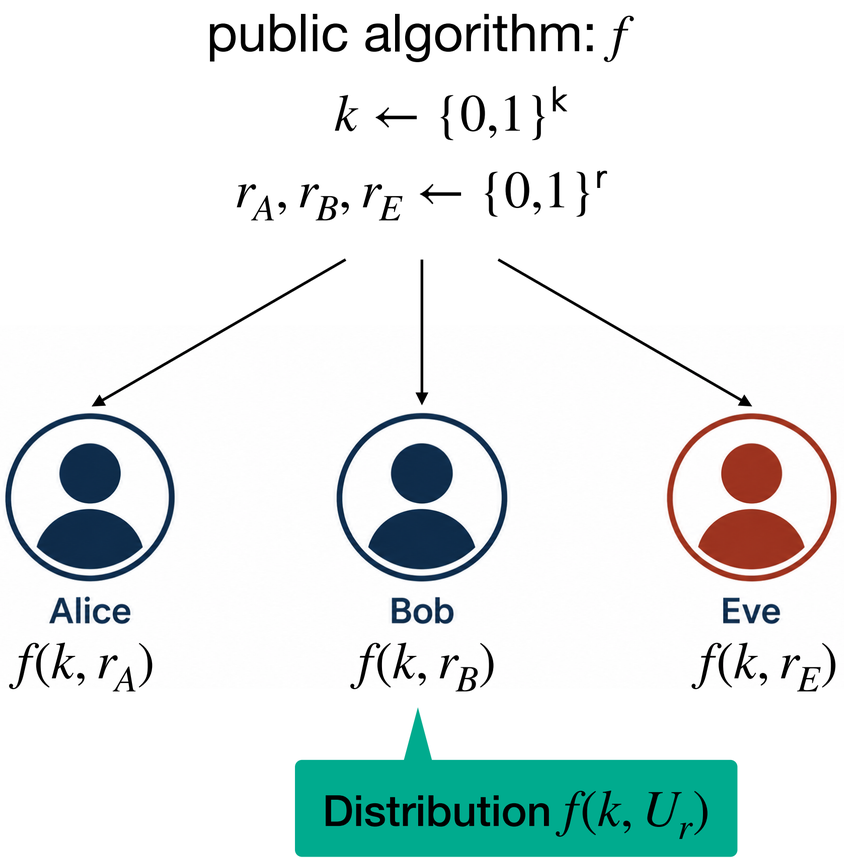
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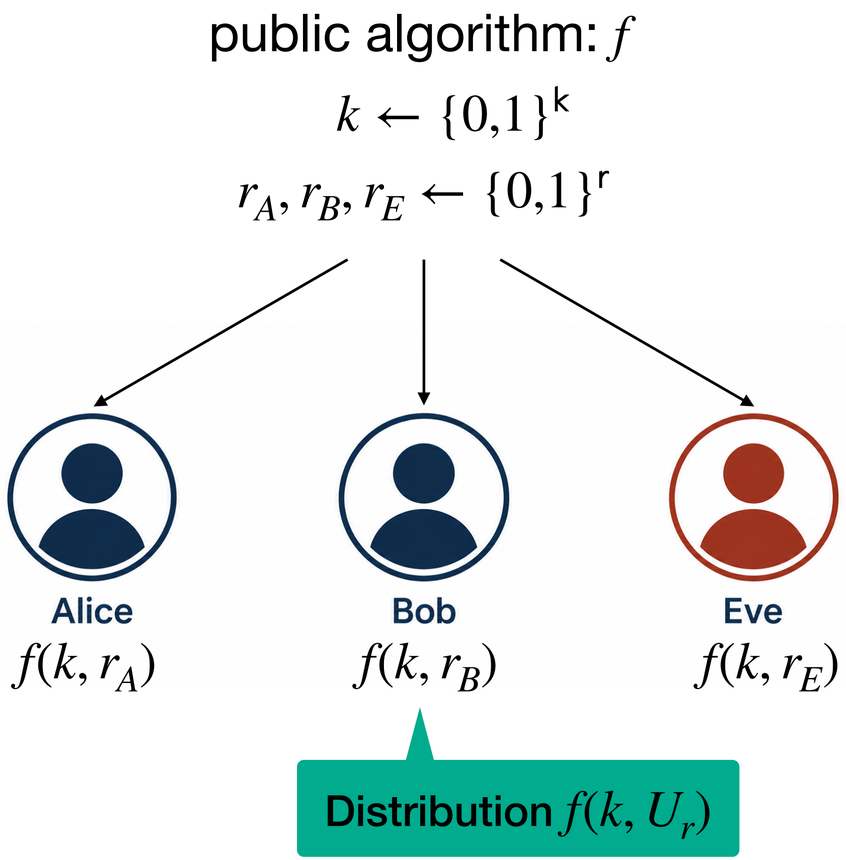


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The common reference distribution model

Compared with CRS

Compared with CRQS



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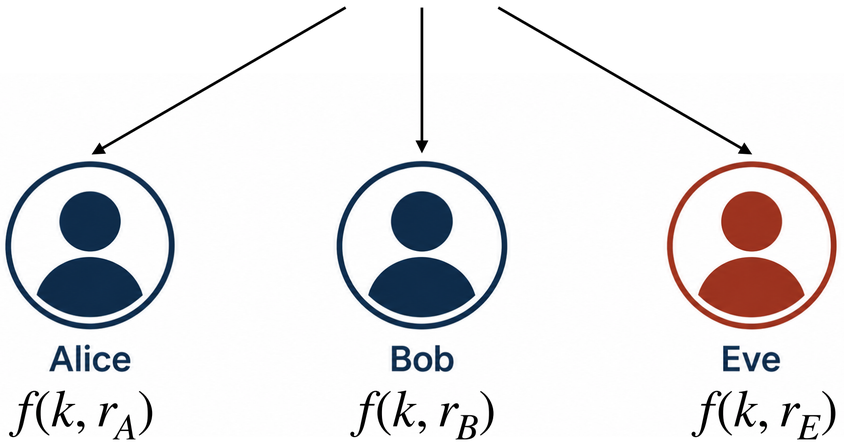
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Distribution $f(k, U_r)$

Compared with CRQS

$k \leftarrow \{0,1\}^{\mathsf{k}}$

$\rho_k = \mathbb{E}_r[\,|f(k,r)\rangle\langle f(k,r)|\,]$

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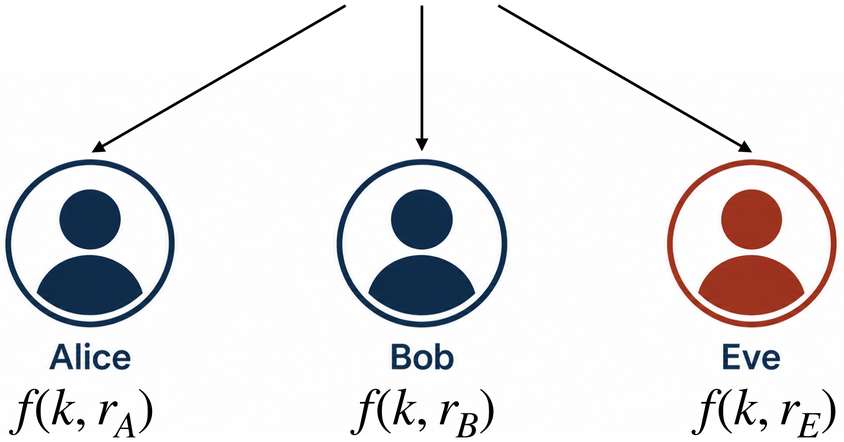
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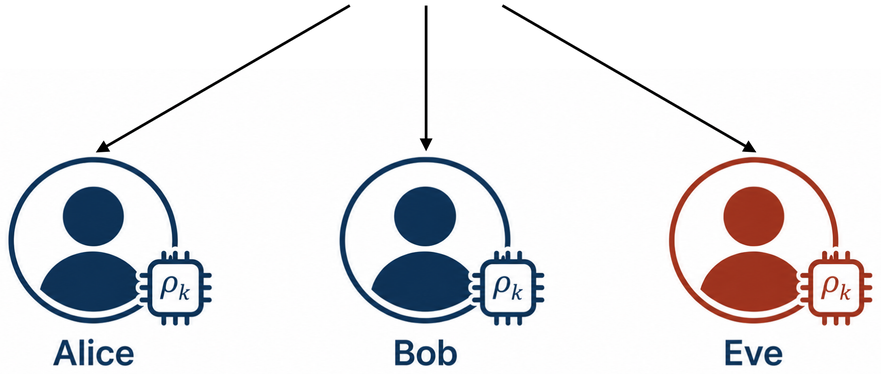


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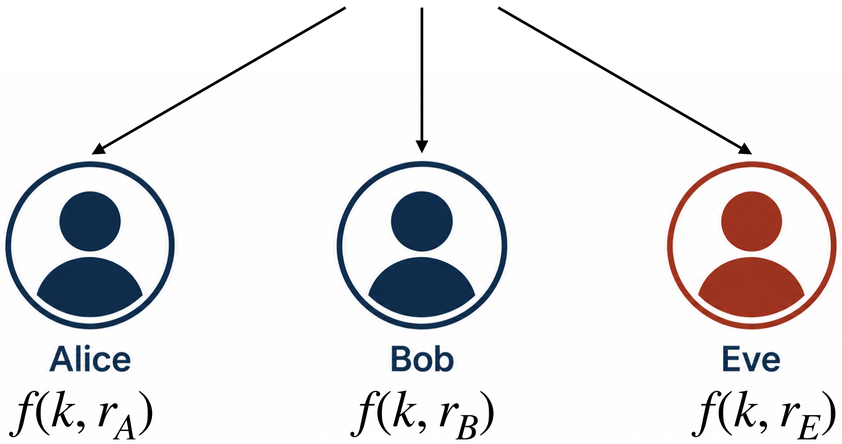
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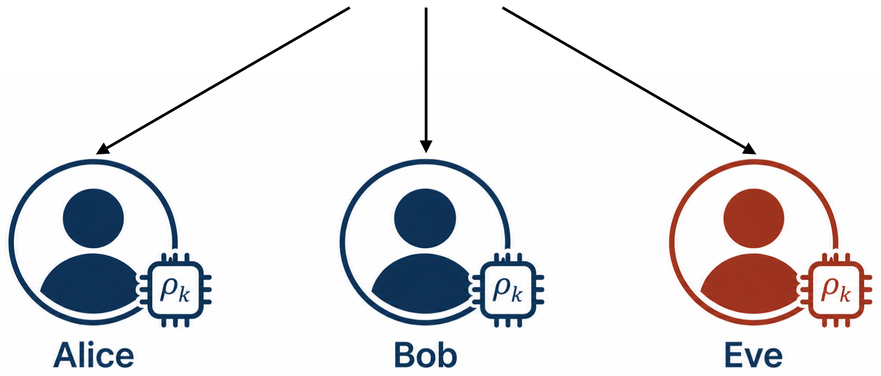
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Our model: CRD

The common reference distribution model

Compared with CRS

CRD seems in between of CRS and CRQS

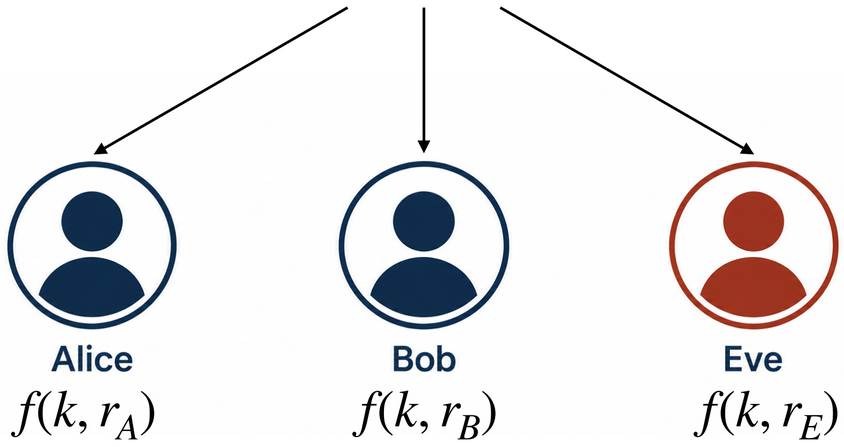


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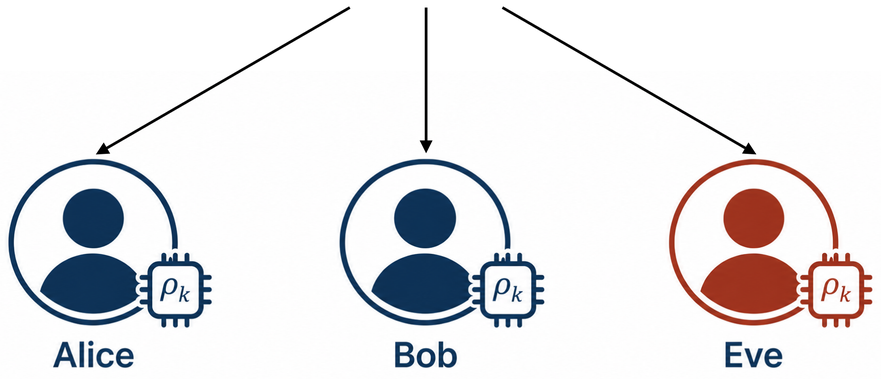
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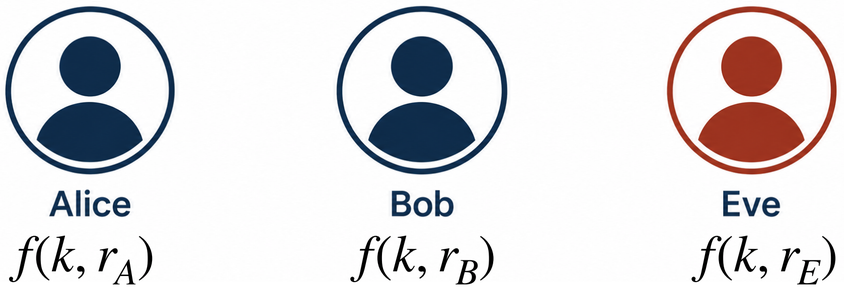
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Distribution $f(k, U_r)$

CRD seems in between of CRS and CRQS 🤔

Not obvious! 😲

=

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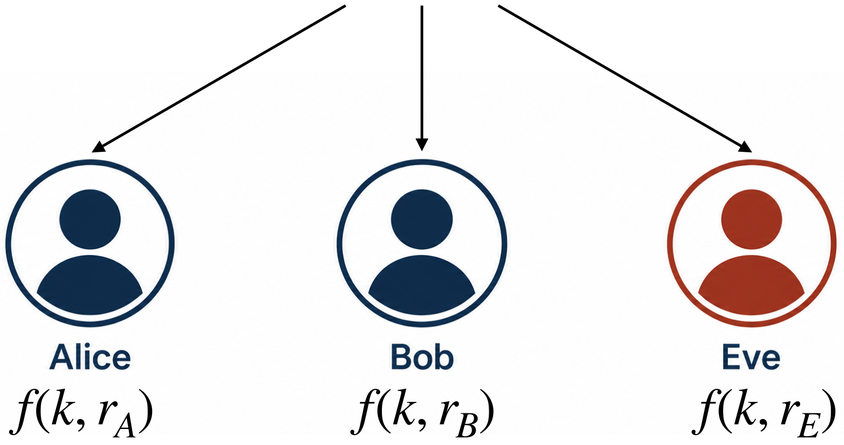
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Compared with CRQS

$k \leftarrow \{0,1\}^k$

Classical mixed state

$\rho_k = \mathbb{E}_r[\,|f(k,r)\rangle\langle f(k,r)|\,]$



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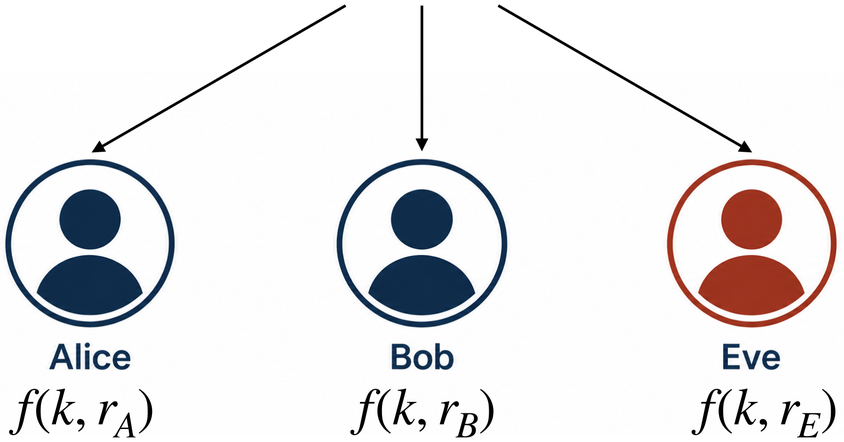
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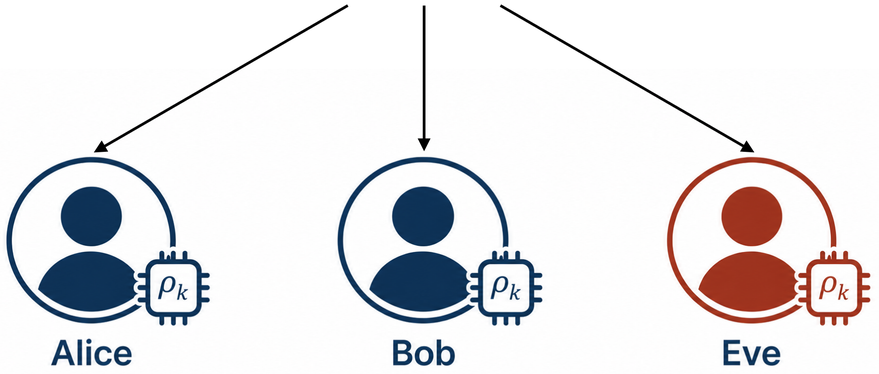


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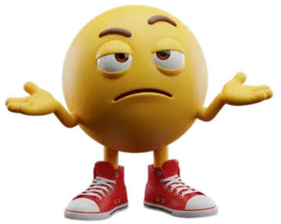
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Classical mixed state

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Why this model?



CRQS is great! But quantum resources is expensive…
Can we build classical primitives?



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Theorem 1. Any protocol in the bCRD model can be unconditionally transformed into an equivalent protocol in the bCRQS model, preserving classical communication. The same holds in the unbounded setting assuming one-way functions (OWFs).

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Not obvious: CRD sample is a mixed state;
CRQS sample is a pure state.

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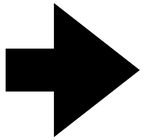
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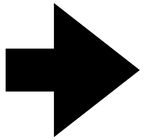
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Classical primitives with unconditional security in ubCRD

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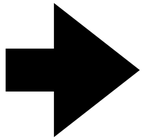
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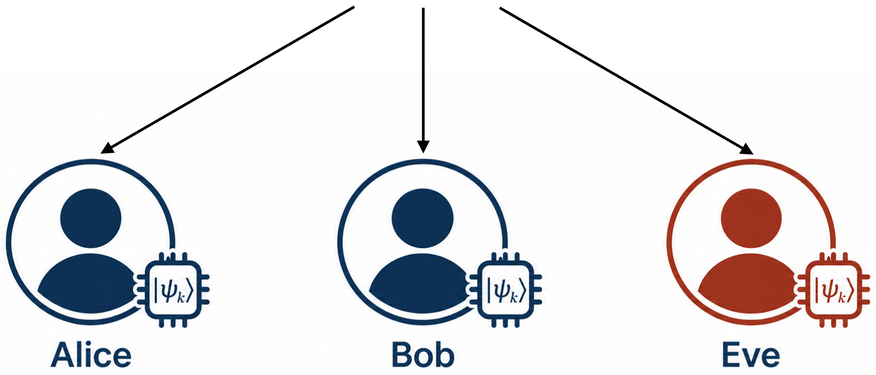
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CRQS

public quantum algorithm: Q

$$k \leftarrow \{0,1\}^{k}$$

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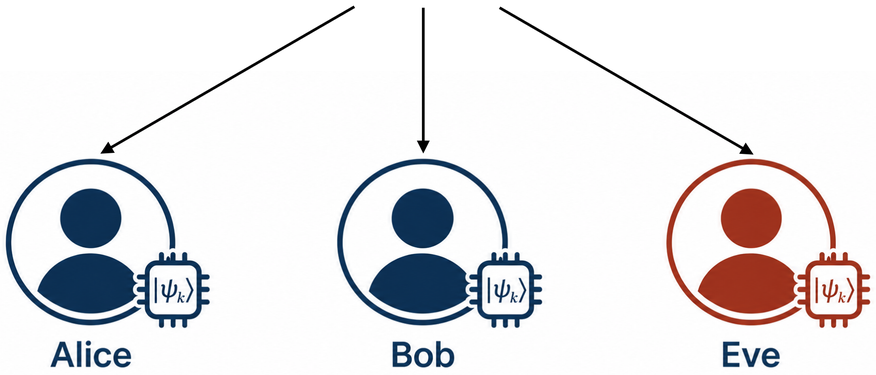


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CHRS

The common Haar random state model

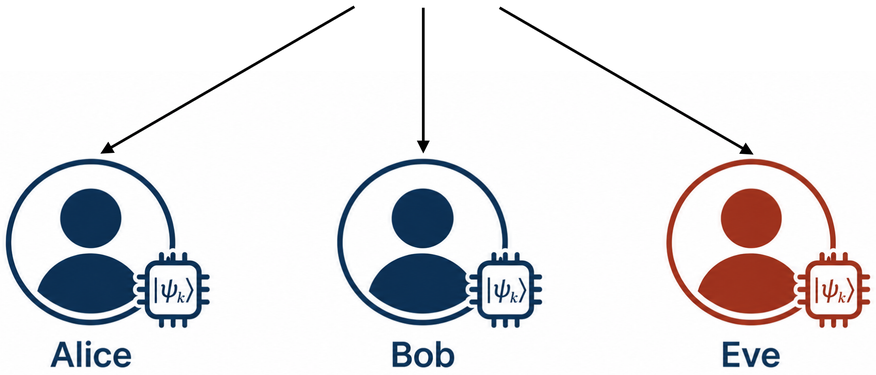
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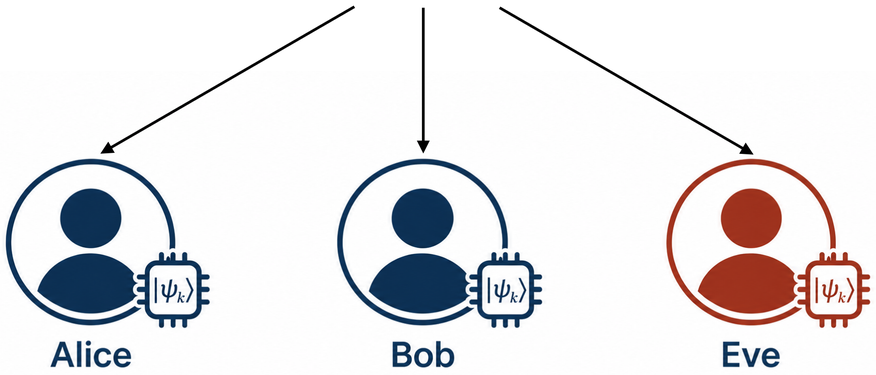
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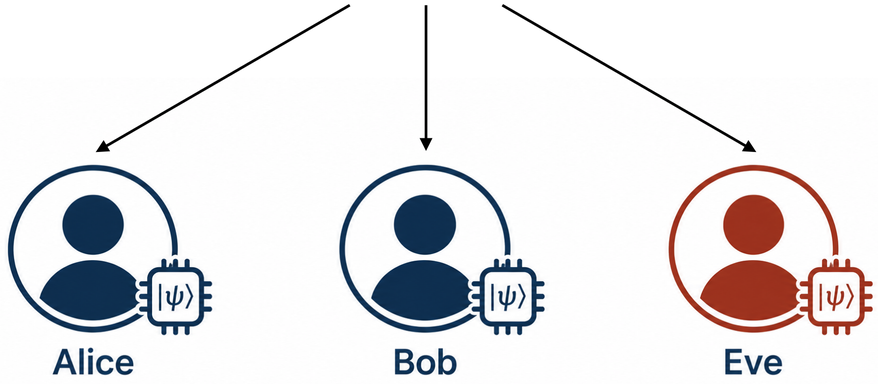
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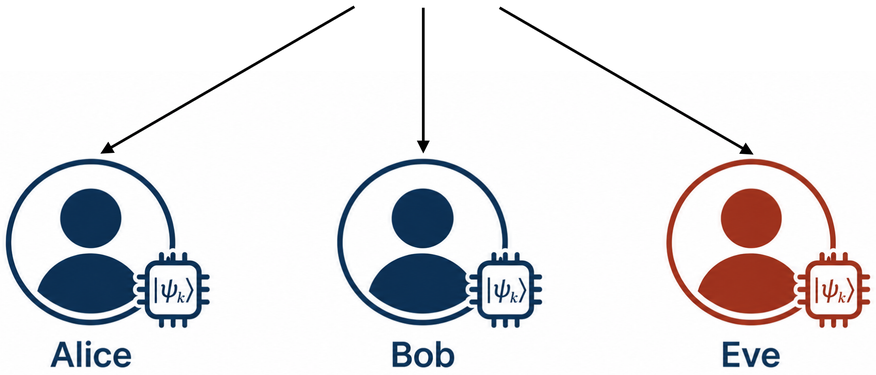
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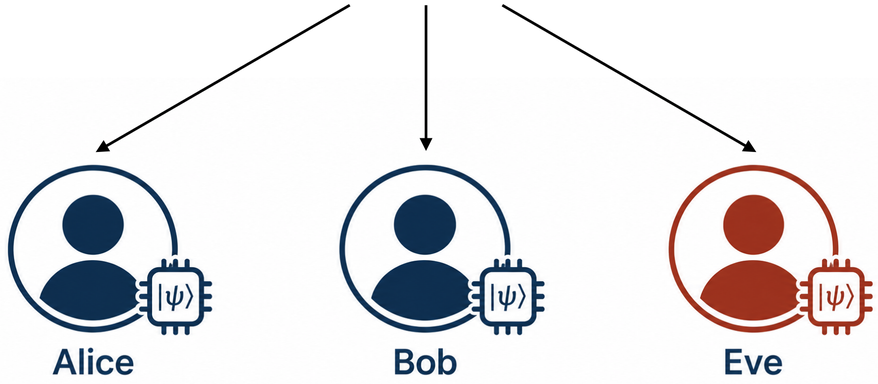


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Not efficient



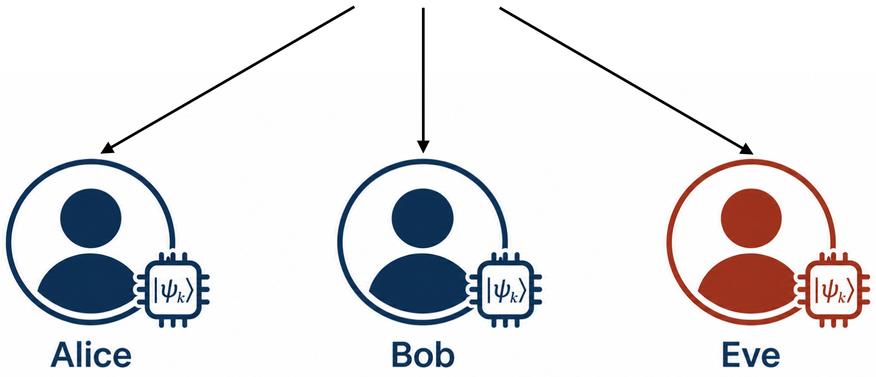
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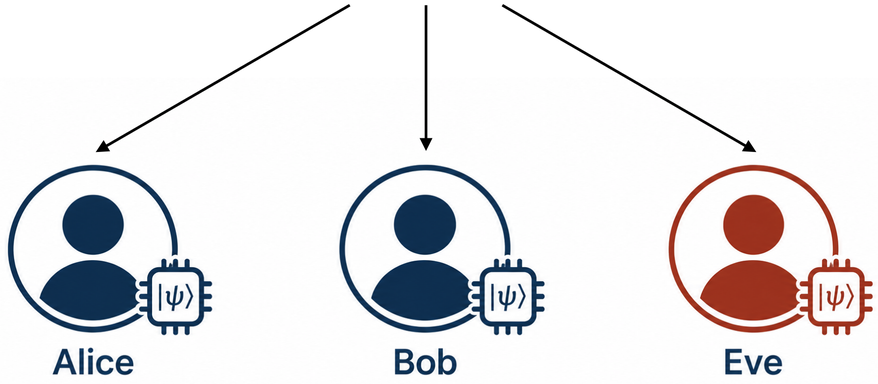
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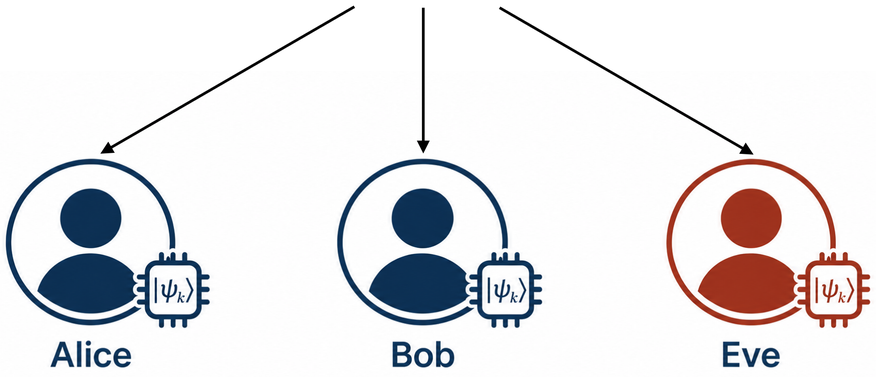
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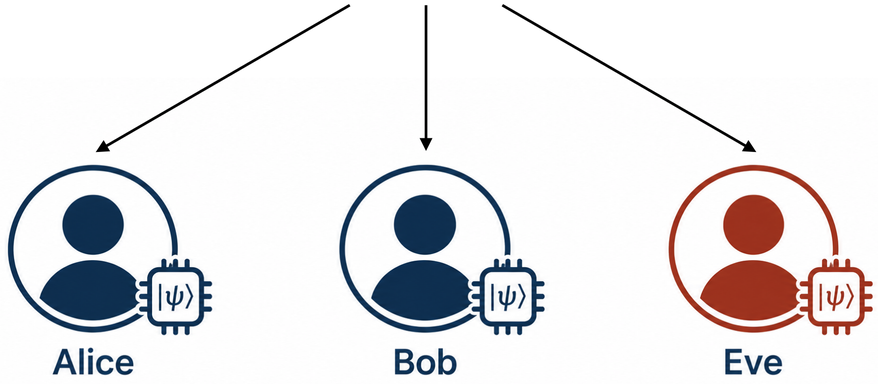
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In unbounded setting:

Statistically-secure quantum commitments

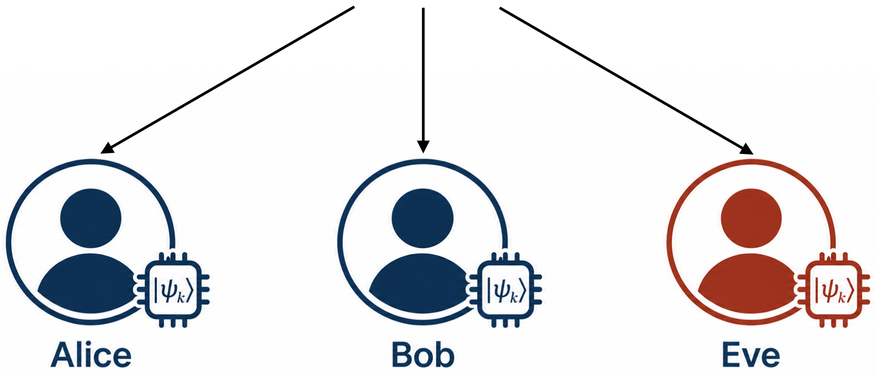
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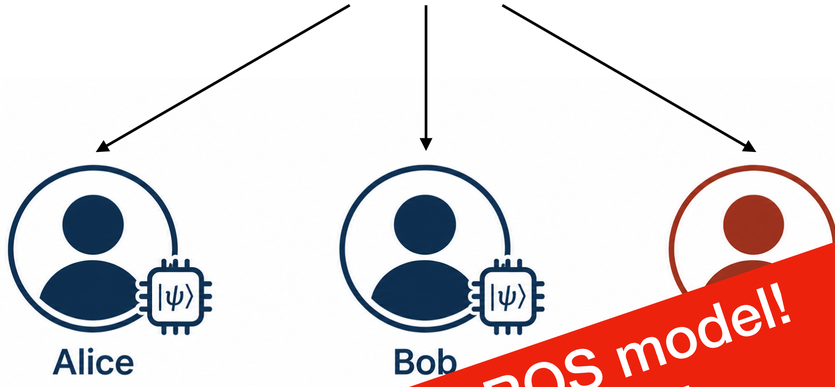
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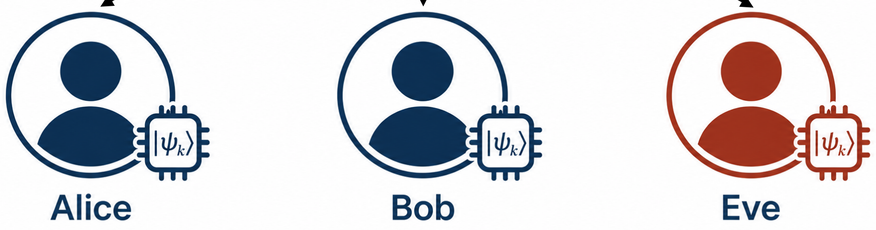
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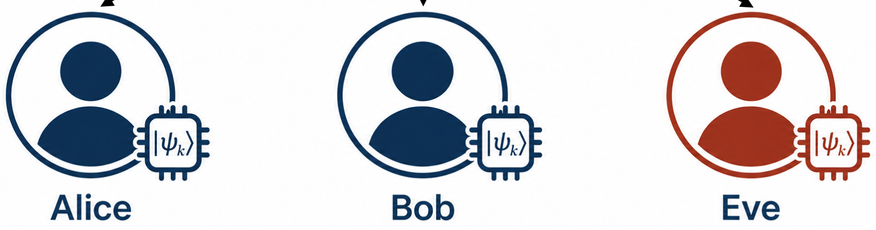
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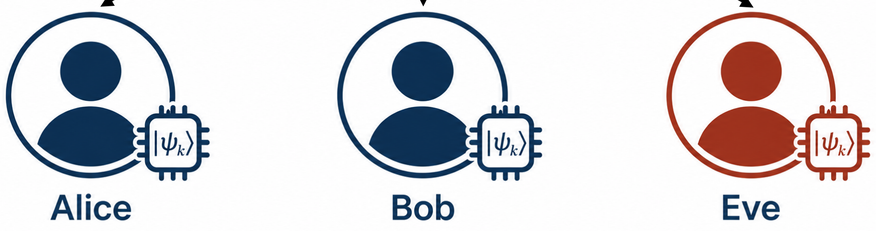
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bCRQS and bCHRS separation

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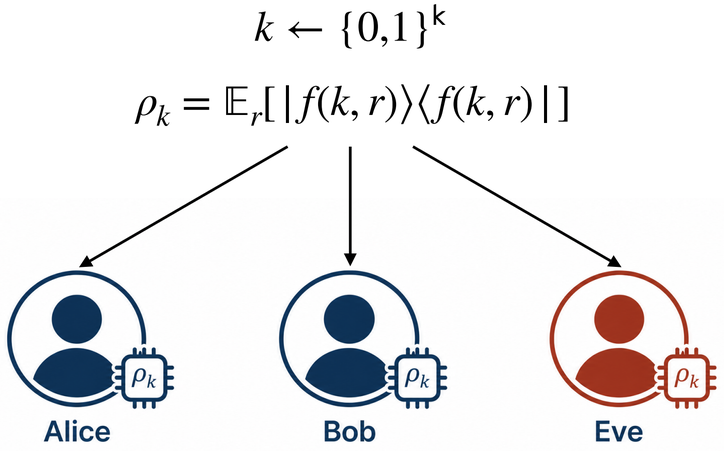
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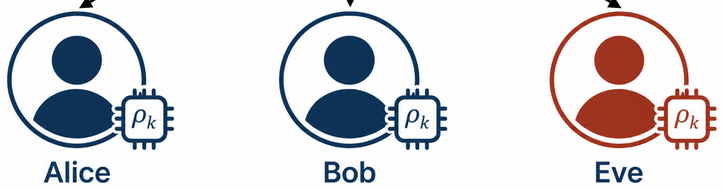


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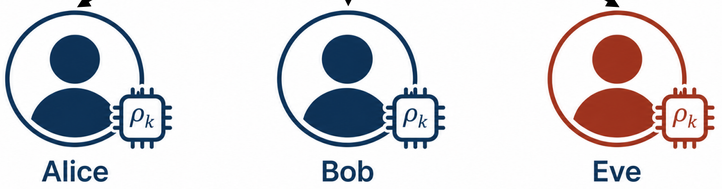


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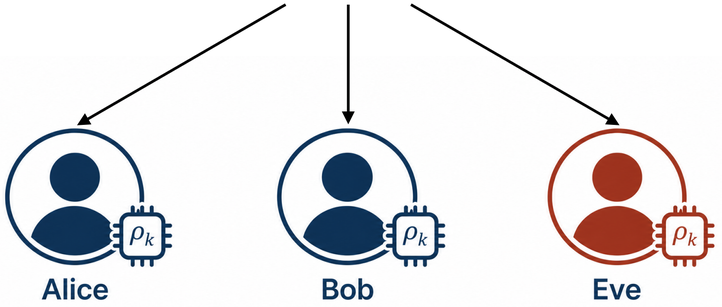
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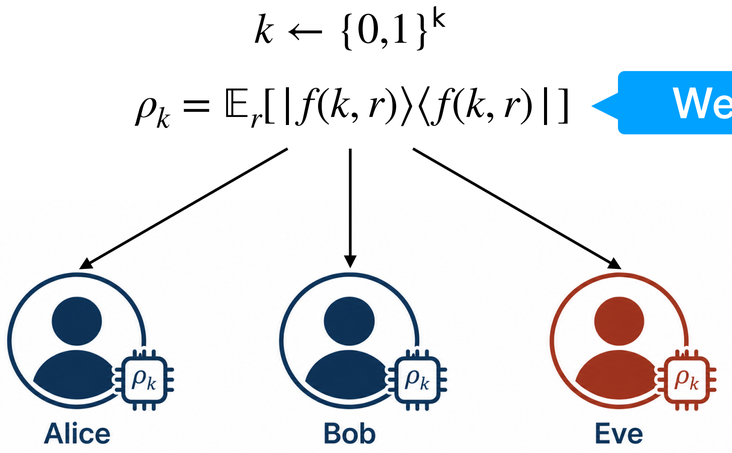


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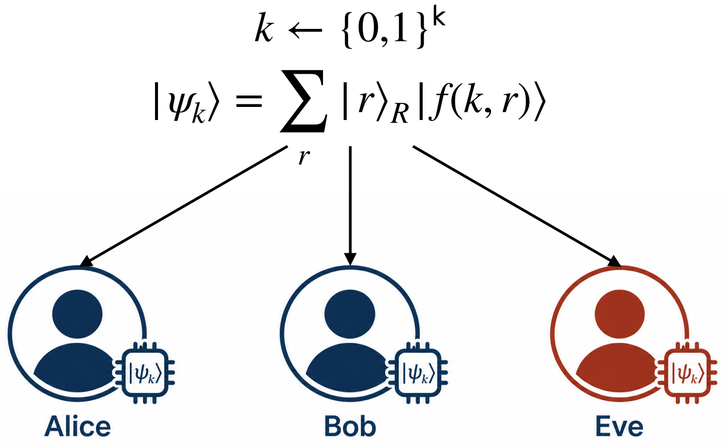
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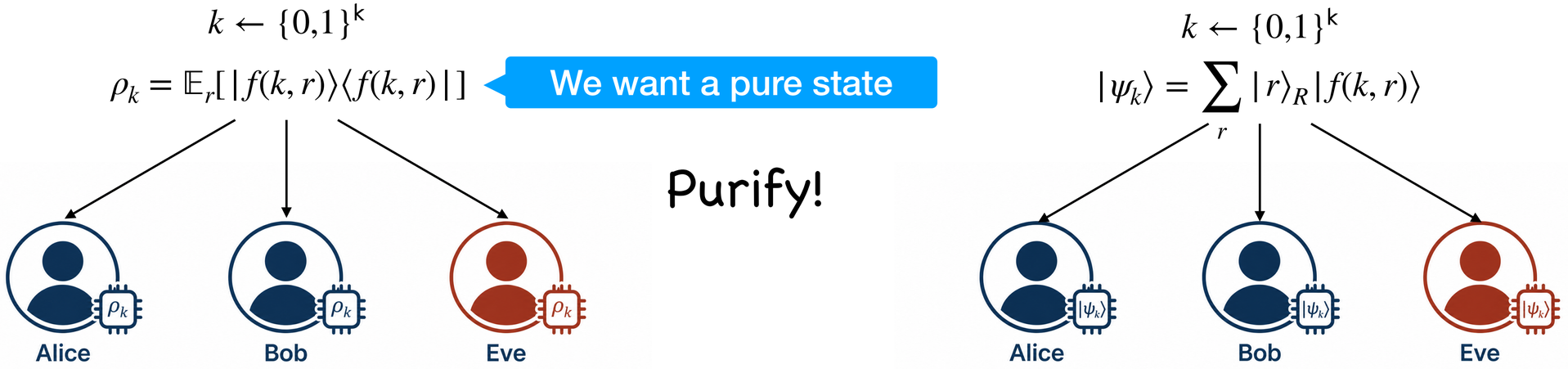


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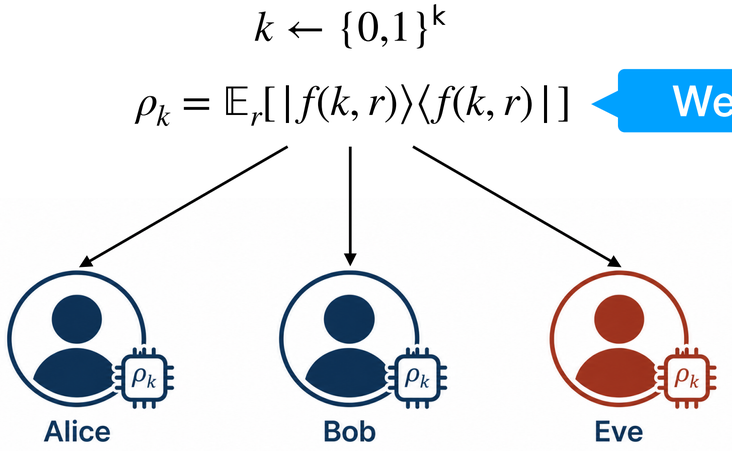


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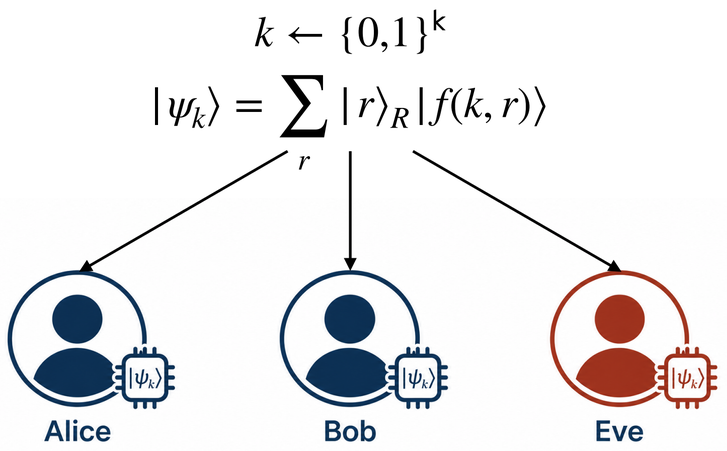
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We want a pure state

Purify!



The security of our construction relies on r remains secret.



Adversary gets the purifying register!

Scramble the secret to hide it!

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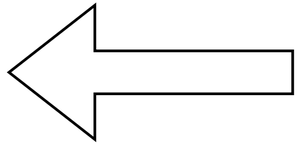
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Coherently permute the pairs

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Instantiate with $2q$-wise independent hash

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$$k \leftarrow \{0,1\}^k$$
$$F \leftarrow \text{Functions}$$
$$|\psi\rangle \leftarrow \text{Haar measure}$$
$$|\psi_k\rangle = U_{k,F}(|\psi\rangle|0\cdots0\rangle)$$

Instantiate with $2q$-wise independent hash

Instantiate with q-design

$$U_{k,F} : |x\rangle|z\rangle \mapsto |x\rangle|z \oplus CRD(k, F(x))\rangle$$

[AKY25]:

$$\mathbb{E}_{\psi \leftarrow Haar}[(|\psi\rangle\langle\psi|)^{\otimes q}] \approx \mathbb{E}_T[|T\rangle\langle T|]$$

Set state $|\{x_1, \ldots, x_q\}\rangle$

$$\mathbb{E}_{\psi,F}[(|\psi_k\rangle\langle\psi_k|)^{\otimes q}] \approx \mathbb{E}_{T,F}[U_{k,F}^{\otimes q}|T,0\rangle\langle T,0|(U_{k,F}^{\dagger})^{\otimes q}]$$

Distinct x_i give independent uniform $F(x_i)$

$|\{(x_1, s_1), \ldots, (x_q, s_q)\}\rangle, s_i = CRD(k, F(x_i))$

Does security in bCRD imply security in bCRQS?

$$s_1, \ldots, s_q \xrightarrow{\text{Sample distinct } x_i} |\{(x_1, s_1), \ldots, (x_q, s_q)\}\rangle$$

Coherently permute the pairs

Yes! 🥳

Theorem 2: Constructions

Classical primitives with unconditional statistical security in bCRD

Non-interactive commitments

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CRD key: random subspaces $A_0, A_1 \leq \mathbb{F}_2^m$ of dimension $m/2$

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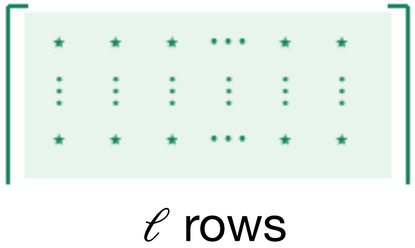
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$m/2$ columns

Full row rank (w.h.p.)



ℓ rows

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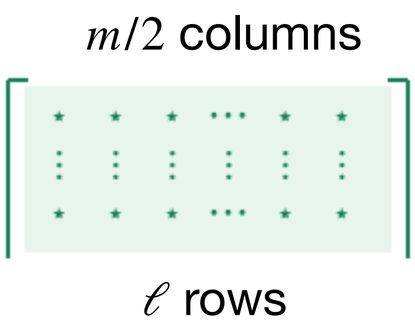
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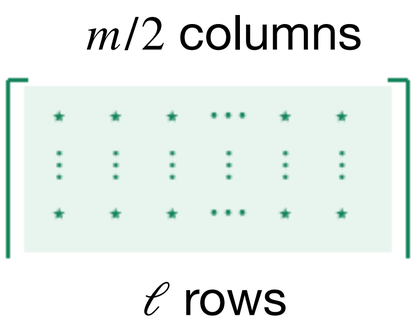
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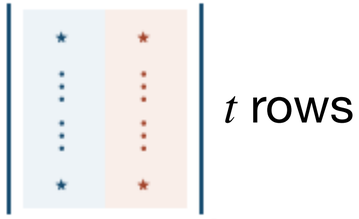
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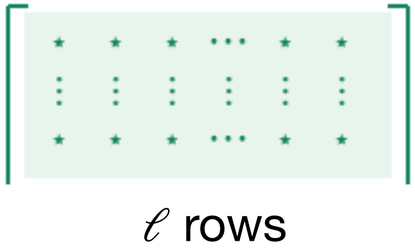
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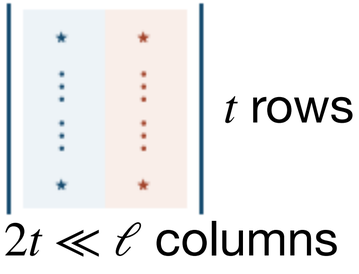
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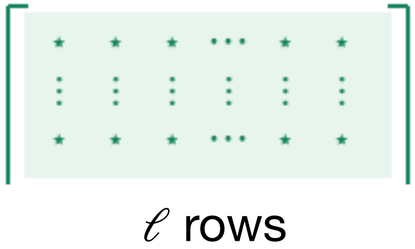
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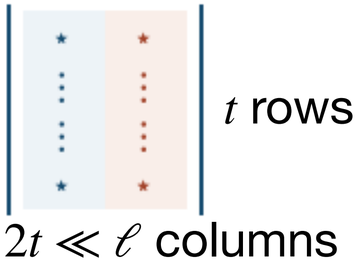
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No double opening for one commitment.

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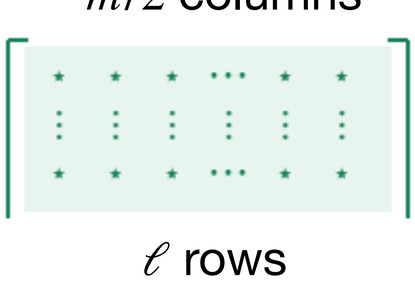
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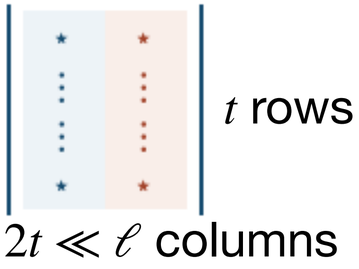
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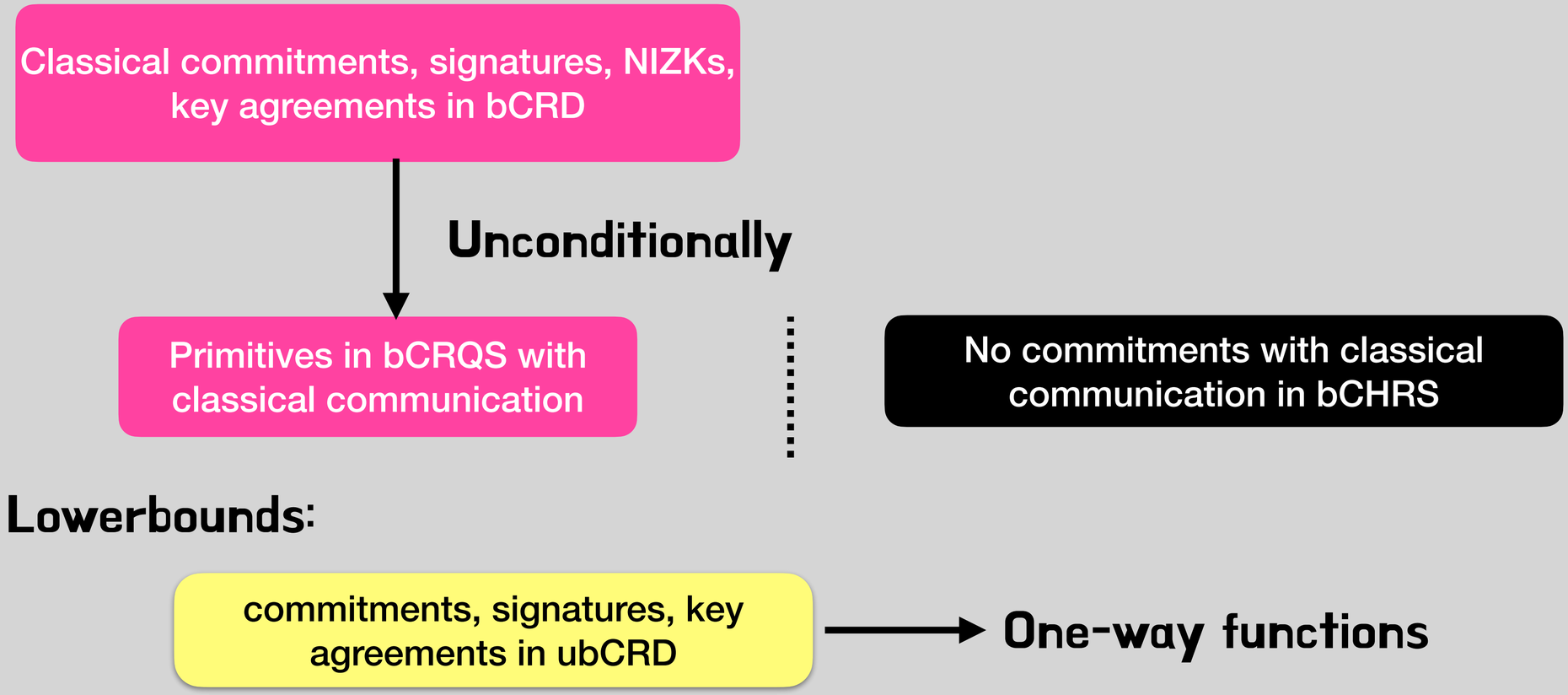


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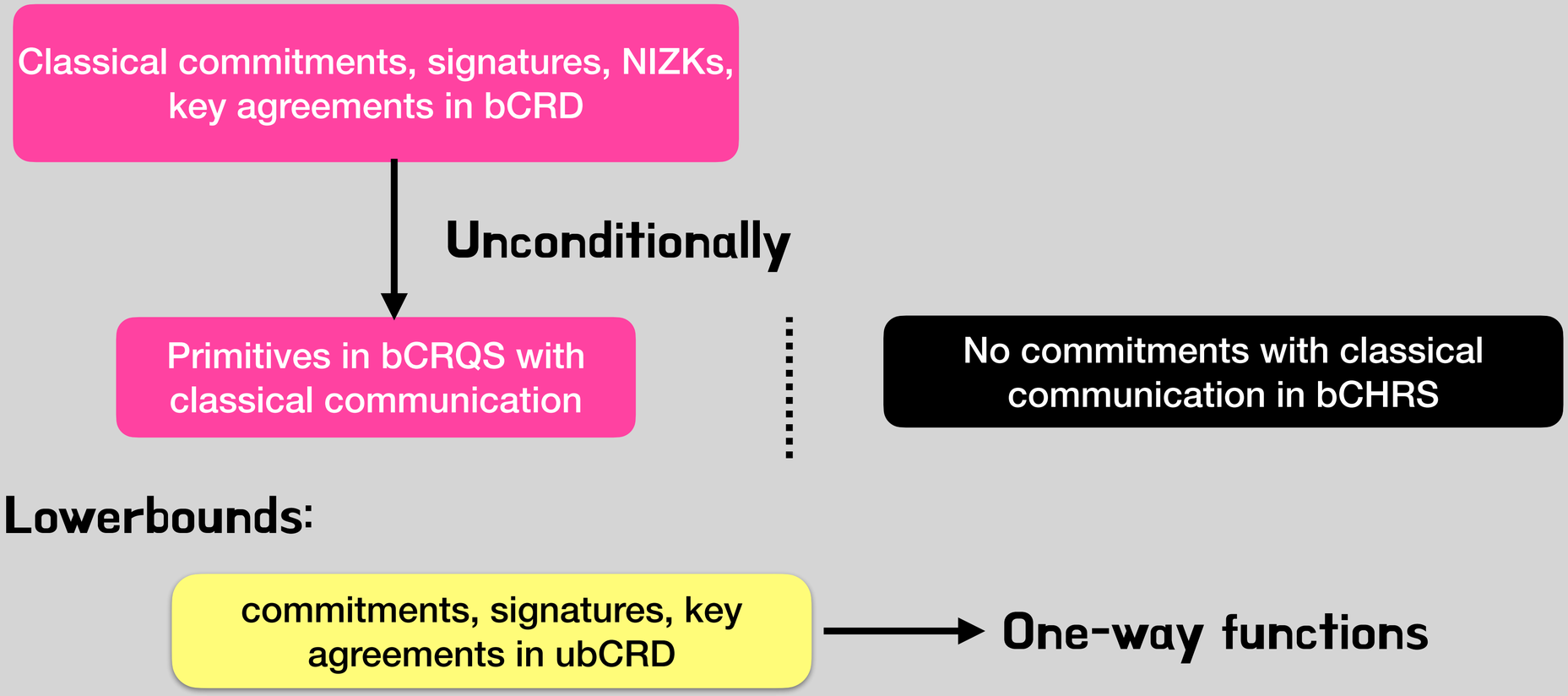
No double opening for one commitment.

Fun! You should also try!

Takeaways


Classical commitments, signatures, NIZKs, key agreements in bCRD
Unconditionally
Primitives in bCRQS with classical communication
No commitments with classical communication in bCHRS
Lowerbounds:
commitments, signatures, key agreements in ubCRD
One-way functions

Takeaways



Thank you!